

Supplementary Information: Payload-Aware Autonomous ADC Linker Design

Elia Savino^{1,d}, Joshua W. Sin^{2,3,d}, Morgan G. L. Reigner^{1,d}, Miquèl À. Pérez-Puigdomènech^{2,d}, Derk H. W. t

1 Full context ranking (all 38 contexts)

Handle	Trigger	Payload	mean	best	SD	seeds
Maleimide	Val-Cit-PABC	cytotoxin	0.88	0.88	0.00	1
DBCO	Non-cleavable-rigid	oligonucleotide	0.88	0.88	0.00	1
DBCO	Non-cleavable	oligonucleotide	0.88	0.88	0.00	1
Maleimide	Glucuronide	cytotoxin	0.45	0.45	0.00	1
Maleimide	Non-cleavable	cytotoxin	0.45	0.45	0.00	1
Bromoacetamide	Val-Cit-PABC	cytotoxin	0.45	0.45	0.00	1
Bromoacetamide	Glucuronide	cytotoxin	0.45	0.45	0.00	1
Bromoacetamide	Non-cleavable	cytotoxin	0.45	0.45	0.00	1
DBCO	Val-Cit-PABC	cytotoxin	0.45	0.45	0.00	1
DBCO	Glucuronide	cytotoxin	0.45	0.45	0.00	1
DBCO	Non-cleavable	cytotoxin	0.45	0.45	0.00	1
Disulfide	Val-Cit-PABC	cytotoxin	0.45	0.45	0.00	1
Disulfide	Glucuronide	cytotoxin	0.45	0.45	0.00	1
Disulfide	Non-cleavable	cytotoxin	0.45	0.45	0.00	1
NHS_ester	Val-Cit-PABC	cytotoxin	0.45	0.45	0.00	1
NHS_ester	Glucuronide	cytotoxin	0.45	0.45	0.00	1
NHS_ester	Non-cleavable	cytotoxin	0.45	0.45	0.00	1
Oxamine	Val-Cit-PABC	cytotoxin	0.45	0.45	0.00	1
Oxamine	Glucuronide	cytotoxin	0.45	0.45	0.00	1
Oxamine	Non-cleavable	cytotoxin	0.45	0.45	0.00	1
Tetrazine	Val-Cit-PABC	cytotoxin	0.45	0.45	0.00	1
Tetrazine	Glucuronide	cytotoxin	0.45	0.45	0.00	1
Tetrazine	Non-cleavable	cytotoxin	0.45	0.45	0.00	1
TCO	Val-Cit-PABC	cytotoxin	0.45	0.45	0.00	1
TCO	Glucuronide	cytotoxin	0.45	0.45	0.00	1
TCO	Non-cleavable	cytotoxin	0.45	0.45	0.00	1
Maleimide	Val-Cit-PABC	oligonucleotide	0.45	0.45	0.00	1
Maleimide	Val-Cit-PABC	immunomodulator	0.45	0.45	0.00	1
Maleimide	Non-cleavable-rigid	cytotoxin	0.45	0.45	0.00	1
Maleimide	Non-cleavable-rigid	oligonucleotide	0.45	0.45	0.00	1
Maleimide	Non-cleavable-rigid	immunomodulator	0.45	0.45	0.00	1
Maleimide	Non-cleavable	oligonucleotide	0.45	0.45	0.00	1
Maleimide	Non-cleavable	immunomodulator	0.45	0.45	0.00	1
DBCO	Val-Cit-PABC	oligonucleotide	0.45	0.45	0.00	1
DBCO	Val-Cit-PABC	immunomodulator	0.45	0.45	0.00	1
DBCO	Non-cleavable-rigid	cytotoxin	0.45	0.45	0.00	1
DBCO	Non-cleavable-rigid	immunomodulator	0.45	0.45	0.00	1
DBCO	Non-cleavable	immunomodulator	0.45	0.45	0.00	1

2 Payload-class re-scoring (mean composite by payload rule)

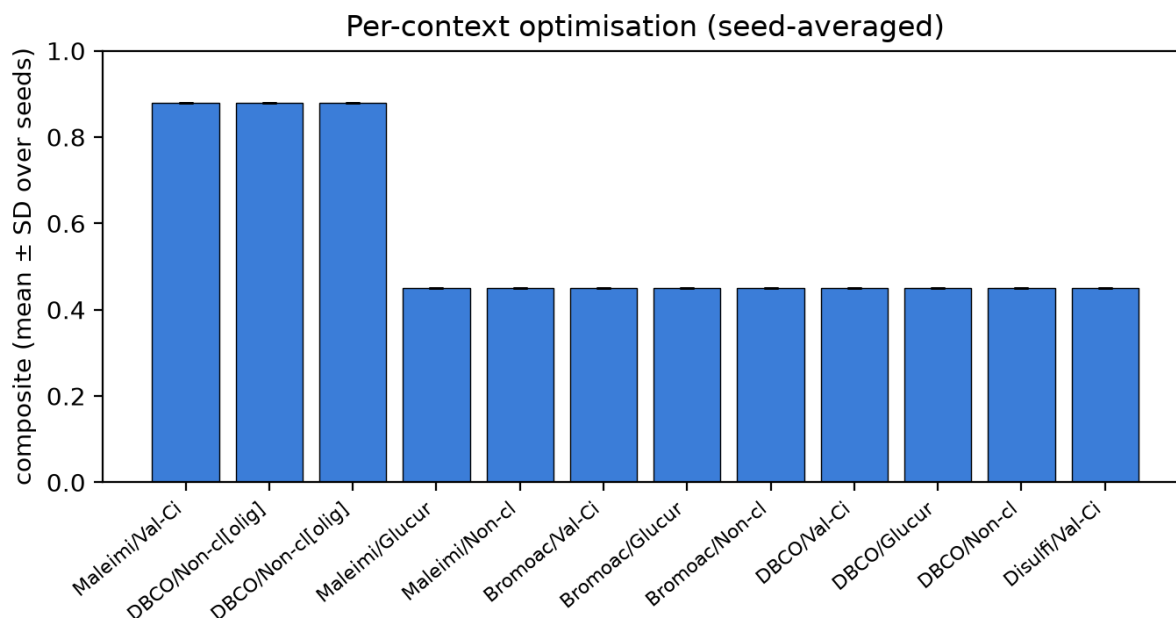


Figure 1: Per-context optimisation, mean $\pm$ SD over seeds (top contexts). Modest seed variance supports the ranking.

Handle	Trigger	cytotoxin	oligonucleotide	immunomodulator
DBCO	Non-cleavable-rigid	0.45	0.88	0.45
DBCO	Non-cleavable	0.45	0.88	0.45
Maleimide	Val-Cit-PABC	0.88	0.45	0.45
Maleimide	Non-cleavable-rigid	0.45	0.45	0.45
Maleimide	Non-cleavable	0.45	0.45	0.45
DBCO	Val-Cit-PABC	0.45	0.45	0.45

3 Table S1: Stability-surrogate calibration

Mechanistic class	stability	expected
acid-labile hydrazone	0.00	low
redox disulfide	1.00	intermediate
protease dipeptide	1.00	high
non-cleavable thioether	1.00	high

4 Commercial linker benchmark

Name	ADC / class	composite	QED	MW
mc-Val-Cit-PABC	brentuximab vedotin (Ad...	0.52	0.116	573
mc-Val-Ala-PABC	Val-Ala class	0.59	0.244	487
mc-GGFG	trastuzumab deruxtecan ...	0.58	0.128	530
MCC (non-cleavable)	trastuzumab emtansine (...)	0.67	0.737	237
SPDB disulfide	mirvetuximab soravtansi...	0.74	0.600	229
sulfo-SPDB	hindered disulfide class	0.76	0.444	309
AcBut hydrazone	gemtuzumab ozogamicin (...)	0.00	0.340	236
beta-glucuronide	glucuronide linker class	0.00	0.156	508
maleimidocaproyl (mc)	mc handle	0.68	0.514	211

5 Table S2: Top designed linkers (full SMILES)

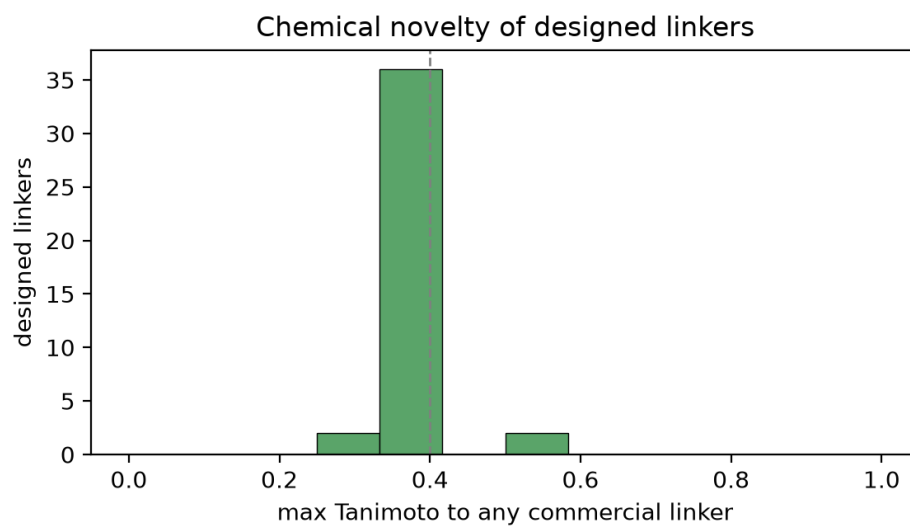


Figure 2: Chemical novelty: nearest-neighbour Tanimoto of each designed linker to any commercial linker. Most designs are distinct chemotypes (similarity < 0.4), not rediscoveries.